THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Full Syllabus (Chapters 1–10) | Class 6 Maths | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(c)** 105	Complete graph on n points: $\frac{n(n-1)}{2}$.
2	**(a)** 150	Find the new time first, then the angle: each hour mark is 30°.
3	**(b)** 5	Repeat the largest-minus-smallest step.
4	**(b)** 140	Total minus the largest.
5	**(d)** 1001	Divide 1000 by the number and round up.
6	**(d)** 162	Let breadth $= b$.
7	**(a)** $\frac{23}{24}$	Use the LCM of the denominators.
8	**(b)** 2	
9	**(c)** 5	Number of sides $= 360 \div a$.
10	**(d)** 22	Simplify step by step.
11	**(c)** 37	Try triangular numbers.
12	**(a)** 1170	$360n + 90$.
13	**(d)** 24	List the sequence.
14	**(d)** 4	Divide by 3 (the symbols are in the same proportion).
15	**(b)** 19	Total numbers − primes − 1 (1 is neither).
16	**(b)** 16	Perimeters are equal.
17	**(a)** Both A and R are true and R is the correct explanation of A.	15 + 4 = 19, and R is the rule of the pattern.
18	**(a)** Both A and R are true and R is the correct explanation of A.	135 lies between 90 and 180, and R is exactly the definition.
19	**(a)** Both A and R are true and R is the correct explanation of A.	7, 5, 3, 0 in descending order gives 7530.
20	**(a)** Both A and R are true and R is the correct explanation of A.	3.5 x 10 = 35, and R says what the half symbol means.

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	Complete graph idea: each pair plays once (1); $\frac{14\times13}{2} = 91$ (1).	1+1
22	Hour hand points to 4 on the dial; smaller angle $= 120^\circ$ (1); Hour hand turns $2 \times 30 = 60^\circ$ in 2 hours (1).	1+1
23	Largest minus smallest each time; $1234 \to 3087 \to 8352 \to 6174$ (1); Steps $= 3$ (1).	1+1
24	Milk $= 150 - 62 - 48$ (1); 40 people (1).	1+1
25	LCM of 8, 12, 18 (1); $= 72$ (1).	1+1
26	$l + b = 30$; ratio parts $= 5$ (1); Length $= 18$ cm, breadth $= 12$ cm (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	$a + (n-1)d = 365$ gives $n - 1 = (365 - 14) \div 9$ (1); $n = 40$ (1); Term before it $= 356$ (1).	1+1+1
28	Hour hand points to 1 on the dial; smaller angle $= 30^\circ$ (1); Larger angle $= 360 - 30 = 330^\circ$ (1); Reflex angle: the larger angle 330° (1).	1+1+1
29	Number 1551 $= 1551$; $1551 \div 11 = 141$ exactly (1); Digit sum $= 12$ (1); Palindrome reads the same backwards (1).	1+1+1
30	Week 1 total $= 61$; Week 2 total $= 60$ (1); Larger: Week 1 (1); Difference $= 1$ (1).	1+1+1
31	LCM of 48, 72, 108 is 432 seconds (1); That is 7 minutes 12 seconds (1); Time: 8:07:12 am (1).	1+1+1
	OR: Perimeter $= 130$ m; Wire needed $= 3 \times 130 = 390$ m (1); Fencing cost $= 4680$ (1); Ploughing cost $= 4000$ (1).	1+1+1
32	Girls =\frac{5}{8}$\times 480 = 300$ (1); Chess girls =\frac{2}{5}$\times 300 = 120$ (1); Boys $= 480 - 300 = 180$ (1).	1+1+1

	OR: Distance between adjacent centres $= 2r = 6$ cm (1); Side $= 6$ cm (1); Area $= 36$ cm^2 (1).	1+1+1

SECTION — D (Long Answer) — Answers

6 × 5 = 30

Q.	Expected Answer / Value Points	Marks
33	(i) 13 sticks (1). (ii) 3 sticks (1). (iii) $4 + 9 \times 3 = 31$ sticks (1). (iv) $40 - 4 = 36$ and $36 \div 3 = 12$ more squares, so $1 + 12 = 13$ squares (2).	1+1+1+2
34	(i) $20 \times 6 = 120°$ (1). (ii) $30°$ (1). (iii) At 4 o'clock the angle is $4 \times 30 = 120°$, an obtuse angle (1). (iv) At 7 o'clock one angle is $210°$ and the smaller angle is $360° - 210° = 150°$ (1); the $210°$ angle is a reflex angle (1).	1+1+1+2
35	Number 12321 (1); Digit sum 9 (1); Divisible by 9 (1); Smallest number to add: 0 (1); Check: new digit sum is a multiple of 9 (1).	1+1+1+1+1
36	A $= 36$, B $= 72$, C $= 108$ (1); D $= 36 + 72 - 108$ (1); D $= 0$ (1); Height of D $= 0$ cm (2).	1+1+1+2
	OR: (i) Side $= \text{HCF} = 12$ m (1); (ii) Beds $= 5 \times 3 = 15$ (1); (iii) Perimeter of one bed $= 48$ m (1); Total fencing $= 720$ m (1); Cost $= ₹3600$ (1).	1+1+1+1+1
37	Rectangle area $= 240$ cm^2 (1); Triangle area $= \frac{1}{2} \times 20 \times 8 = 80$ cm^2 (1); Remaining $= 160$ cm^2 (1); Fraction left $= \frac{160}{240}$ (2).	1+1+1+2
	OR: After day 1: \frac{2}{3} left (1); Day 2 sells $\frac{1}{6}$ of total (1); Left fraction $\frac{1}{2}$ (1); Total $= 30 \div \frac{1}{2} = 60$ m (2).	1+1+1+2
38	Draw AB = 6 cm (1); With A as centre and radius 10 cm draw an arc; mark the corner C using the perpendicular at B (1); Other side BC = 8 cm (1); Half diagonal OA $= 5$ cm (1); Perimeter $= 2(6 + 8) = 28$ cm (1).	1+1+1+1+1
	OR: (a) 6 (1); (b) $60°$ (1); (c) 30 cm (1); (d) $240°$ (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Full Syllabus (Chapters 1–10) | Class 6 Maths | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(a)** 800	Subtract the first 10 odd numbers, whose sum is 100.
2	**(c)** 180	Net turn.
3	**(a)** 10	The second digit can be any of 10 digits.
4	**(b)** 84	Multiply.
5	**(d)** 24	LCM.
6	**(b)** 120	Half of the rectangle.
7	**(b)** 40	Fraction removed = difference.
8	**(b)** 36	Side $= 2r$.
9	**(d)** 2	Letters H, I, O, X.
10	**(c)** 23	Distance $= \|a - b\|$.
11	**(d)** 39	$20^2 - 19^2$ is the next odd number.
12	**(b)** 30	The hands are placed symmetrically about the 12-6 line, so the angle is the same.
13	**(a)** 165	Add the numbers.
14	**(d)** 9	Add the children, then divide by the value of one symbol.
15	**(c)** 24	HCF of three numbers.
16	**(c)** 336	Outer minus inner.
17	**(a)** Both A and R are true and R is the correct explanation of A.	A prime has exactly two factors and a composite has more than two; 1 has just one factor.
18	**(a)** Both A and R are true and R is the correct explanation of A.	2 x (12 + 5) = 34.
19	**(a)** Both A and R are true and R is the correct explanation of A.	Multiplying 3 and 4 by 2 gives 6 and 8.
20	**(a)** Both A and R are true and R is the correct explanation of A.	Every chord through the centre has length 2r, and no other chord is longer.

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	Number $= 18 \div$ \frac{3}{4}$= 24$ (1); $\frac{1}{3}$ of the number $= 8$ (1).	1+1
22	Inside: 2; On: 1 (1); Outside: 1 (1).	1+1
23	Sides $= 360 \div 30 = 12$ (1); Perimeter $= 12 \times 4 = 48$ cm (1).	1+1
24	Opposite integers cancel each other; Remaining: integers from 7 to 10 (1); Sum $= 34$ (1).	1+1
25	$n \times n = 961$ so $n = 31$ (1); 31th odd number $= 2 \times 31 - 1 = 61$ (1).	1+1
26	Total $= 360 \times 2 + 90 = 810°$ (1); Right angles $= (810) \div 90 = 9$ (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	Angle $= 360 \div 5 = 72°$ (1); Lines of symmetry $= 5$ (1); Total around centre $= 5 \times 72 = 360°$ (1).	1+1+1
28	After dive: -80; After rise: -40 (1); After second dive: -65 (1); Total distance $= 115$ m (1).	1+1+1
29	First 20 odd numbers sum to 400 (1); First 25 odd numbers sum to 625 (1); Required sum $= 225$ (1).	1+1+1
30	Minutes $= 90 \div 6 = 15$ (1); Right angles $= 90 \div 90 = 1$ (1); Time shown: 2:15 (1).	1+1+1
31	Build the row; Row: 2, 3, 5, 8, 13, 21, 34, 55 (1); 8th number $= 55$ (1); Sum $= 141$ (1).	1+1+1
	OR: Pens $= 36$ (1); Notebooks $= 72$ (1); Total $= 108$ (1).	1+1+1
32	Subtract the remainder: 72 and 96 (1); HCF of 72 and 96 is 24 (1); The number is 24 (it must exceed 4) (1).	1+1+1

	OR: Square perimeter $= 4 \times 12 = 48$ cm; rectangle perimeter $= 2(14 + 10) = 48$ cm; Square area $= 144$ cm^2 (1); Rectangle area $= 140$ cm^2 (1); Square is larger by 4 cm^2 (1).	1+1+1

SECTION — D (Long Answer) — Answers

6 × 5 = 30

Q.	Expected Answer / Value Points	Marks
33	(i) 8530 (1). (ii) 3058 (1). (iii) $8530 - 3058 = 5472$ (1). (iv) An odd number must end in 5 or 3. The smallest start is 3 then 0, giving 3058 (even) or 3085; 3085 ends in 5 and is the smallest odd number (2).	1+1+1+2
34	(i) Cricket (1). (ii) $18 + 6 = 24$ children (1). (iii) $18 - 9 = 9$ more (1). (iv) Football, as $12 = 2 \times 6$ (1); Hockey and Kabaddi together: $6 + 9 = 15$ children (1).	1+1+1+2
35	Net $= 180 - 45 + 90 = 225^\circ$ (1); Direction 225° clockwise from North (1); To face North again: turn clockwise 135° more (1); Check: $225 + 135 = 360^\circ$ (2).	1+1+1+2
36	Reverse-and-add repeatedly (1); Sums: 143, 484 (1); Steps $= 2$ (1); Palindrome 484 has 3 digits (odd) (2).	1+1+1+2
	OR: Pink $= 100 - 70 = 30$ (1); Blue fraction $= \frac{25}{100} = \frac{1}{4}$ (2); Symbols: red: 4, blue: 5, green: 3, yellow: 2, pink: 6 (2).	1+2+2
37	LCM $= 24$ (1); Smallest 4-digit multiple: 1008 (1); Largest 4-digit multiple: 9984 (1); Verify divisibility (2).	1+1+1+2
	OR: Path parallel to length: $60 \times 3 = 180$ m^2 (1); Path parallel to breadth: $40 \times 3 = 120$ m^2 (1); Common square $= 3 \times 3 = 9$ m^2 (1); Area of paths $= 291$ m^2 (1); Rest of the park $= 2109$ m^2 (1).	1+1+1+1+1
38	Sum of whole parts $= 6$ (1); Sum of fractions $= 1$ (1); Total $= 7$ (1); After subtracting: $6\frac{1}{4}$ (2).	1+1+1+2
	OR: Row length $= 5 \times 6 = 30$ cm (1); First to last centres $= (5 - 1) \times 6 = 24$ cm (1); Circles in the strip $= 7$ (1); Show the working with diameter (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Full Syllabus (Chapters 1–10) | Class 6 Maths | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(b)** 11	Divide the total increase by the number of steps, which is one less than the term number.
2	**(d)** 25	The minute hand turns 6° per minute.
3	**(d)** 24	Subtract.
4	**(c)** 37	Subtract.
5	**(b)** 5	Digit sum must be a multiple of 9.
6	**(a)** 14	Perimeter does not change.
7	**(a)** 5	Find the number first.
8	**(d)** 16	Each is half the diagonal.
9	**(b)** 3	If a figure has a line and order n, it has n lines.
10	**(b)** -30	Gains positive, losses negative.
11	**(b)** 88	Add the triangular number.
12	**(b)** 4	Each line gives one straight angle.
13	**(b)** 4	Find the palindrome.
14	**(d)** 16	Add the heights.
15	**(d)** 997	Subtract the remainder first.
16	**(d)** 5	Area = cost ÷ rate; breadth = area ÷ length.
17	**(a)** Both A and R are true and R is the correct explanation of A.	A square is a regular polygon with 4 sides.
18	**(a)** Both A and R are true and R is the correct explanation of A.	-7 lies to the left of -3.
19	**(b)** Both A and R are true but R is not the correct explanation of A.	1+3+5+7+9 = 25, and 25 = 5 x 5 is a square, but R does not explain why the sum is 25.
20	**(b)** Both A and R are true but R is not the correct explanation of A.	Both statements are true, but the value of a right angle does not explain the value of a straight angle.

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	Apply the rule (1); Terms: 15, maximum: 52 (1).	1+1
22	First bar = 180; Second bar height = 6 cm (1); Value = 120 (1).	1+1
23	The numbers include a multiple of 3 (1); Hence the product is divisible by 3 (1).	1+1
24	Area = 1200 m^2, perimeter = 140 m; Levelling cost = 6000 (1); Fencing cost = 1680; total = 7680 (1).	1+1
25	Required =\frac{5}{6}−\frac{2}{7} (1); = $\frac{23}{42}$ (1).	1+1
26	$\angle AOB = \angle COD = 35^\circ$ (vertically opposite) (1); $\angle BOC = \angle AOD = 145^\circ$ (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	$\frac{?}{12}$ =\frac{5}{6}−\frac{1}{4} (1); = $\frac{7}{12}$ =\frac{7}{12} (1); Missing numerator = 7 (1).	1+1+1
28	Width of ring = 5 cm (1); Inner diameter = 8 cm (1); Longest chord of outer circle = its diameter = 18 cm (1).	1+1+1
29	Lines through opposite vertices: $4 \div 2 = 2$ (1); Lines through midpoints of opposite sides: 2 (1); Total = 4 (1).	1+1+1
30	Other = $-4 - 11 = -15$ (1); Difference = 26 (1); Check the sum: -4 (1).	1+1+1
31	Row 6 has $1 + 2 + \cdots + 6 = 21$ cubes (1); First rows: 1+3+6+10 (1); Total in first 4 rows = 20 (1).	1+1+1
	OR: Slices = $2 \times 2 = 4$ (1); Angle = $360 \div 4 = 90^\circ$ (1); Type: right (1).	1+1+1
32	Arrangements: $4 \times 3 \times 2 \times 1 = 24$ (1); Largest 9752, smallest 2579 (1); Difference 7173 (1).	1+1+1

	OR: Frequencies: 1: 1; 2: 1; 3: 1; 4: 1; 5: 4; 6: 1; 7: 2; 8: 1 (1); Values greater than 5: 1 + 2 + 1 = 4 (2).	1+2

SECTION — D (Long Answer) — Answers

6 × 5 = 30

Q.	Expected Answer / Value Points	Marks
33	(i) $12 = 2^2 \times 3$, $18 = 2 \times 3^2$ (1). (ii) LCM $= 2^2 \times 3^2 = 36$ (1). (iii) 9:36 am (1). (iv) 11:00 am is 120 minutes after 9:00; $120 \div 36$ gives 3 more times, at 9:36, 10:12 and 10:48 am (2).	1+1+1+2
34	(i) $2 \times (40 + 25) = 130$ m (1). (ii) $40 \times 25 = 1000$ m^2 (1). (iii) $130 \times 20 =$ ₹2600 (1). (iv) $1000 \div 50 = 20$ parts, so $20 \times 2 = 40$ kg (2).	1+1+1+2
35	Figure 15: dots $= 5 + 14 \times 9 = 131$ (1); For 176 dots: steps from figure 1 $= (176 - 5) \div 9 = 19$ (1); Figure number $= 20$ (1); Check: substitute back to get the same number of dots (2).	1+1+1+2
36	(a) Hour hand: 4 hour marks of $30^\circ = 120^\circ$ (1); (b) Minute hand has completed 4 full turns $= 1440^\circ$ and stands at 12 (1); (c) Smaller angle $= 120^\circ$ (1); (d) obtuse angle (2).	1+1+1+2
	OR: Number 24542 (1); Digit sum 17 (1); Not divisible by 9 (1); Smallest number to add: 1 (1); Check: new digit sum is a multiple of 9 (1).	1+1+1+1+1
37	A = 90, B = 60, C = 120 (1); D = 90 + 60 − 120 (1); D = 30 (1); Height of D = 2 cm (2).	1+1+1+2
	OR: (i) Side = HCF = 24 m (1); (ii) Beds = 3 × 2 = 6 (1); (iii) Perimeter of one bed = 96 m (1); Total fencing = 576 m (1); Cost = ₹2304 (1).	1+1+1+1+1
38	Rectangle area = 240 cm^2 (1); Triangle area = $\frac{1}{2} \times 24 \times 6 = 72$ cm^2 (1); Remaining = 168 cm^2 (1); Fraction left = $\frac{168}{240}$ (2).	1+1+1+2
	OR: After day 1: \frac{3}{4} left (1); Day 2 sells $\frac{1}{2}$ of total (1); Left fraction $\frac{1}{4}$ (1); Total = $12 \div \frac{1}{4} = 48$ m (2).	1+1+1+2

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